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APIs > Customer To Business Register URL

Customer To Business Register URL

Register validation and confirmation URLs on M-Pesa

POST `https://sandbox.safaricom.co.ke/mpesa/c2b/v1/registerurl`

Operation: POST

Endpoint: `https://sandbox.safaricom.co.ke/mpesa/c2b/v1/registerurl`

DOCUMENTATION

ERRORS

POSTMAN COLLECTION LINK

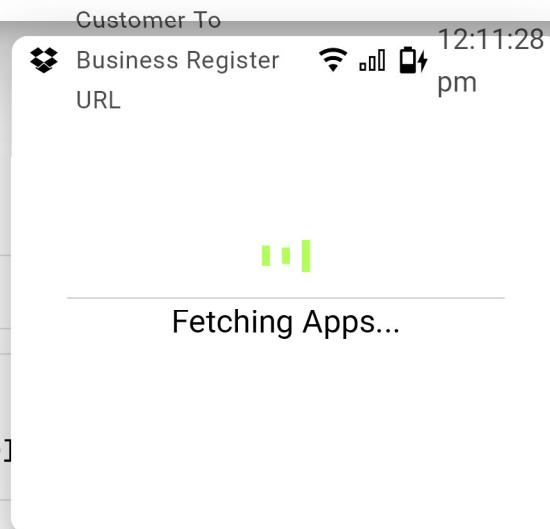
Register URL API works hand in hand with Customer to Business (C2B) APIs and allows receiving payment notifications to your paybill. This API enables you to register the callback URLs via which you shall receive notifications for payments to your pay bill/ till number.

There are two URLs required for Register URL API: **Validation URL** and **Confirmation URL**.

Validation URL: This is the URL that is only used when a Merchant (Partner) requires to validate the details of the payment before accepting. For example, a bank would want to verify if an account number exists in their platform before accepting a payment from the customer.

Confirmation URL: This is the URL that receives payment notification once payment has been completed successfully on M-PESA.

NB: C2B Transaction Validation is an optional feature that needs to be activated on M-Pesa, the owner of the shortcode needs to make this request for activation by emailing us at apisupport@safaricom.co.ke or M-pesabusiness@safaricom.co.ke if they need their transactions validated before execution.





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The above chart, explains how URLs are registered to send requests and receive responses

Flowchart of the C2B process

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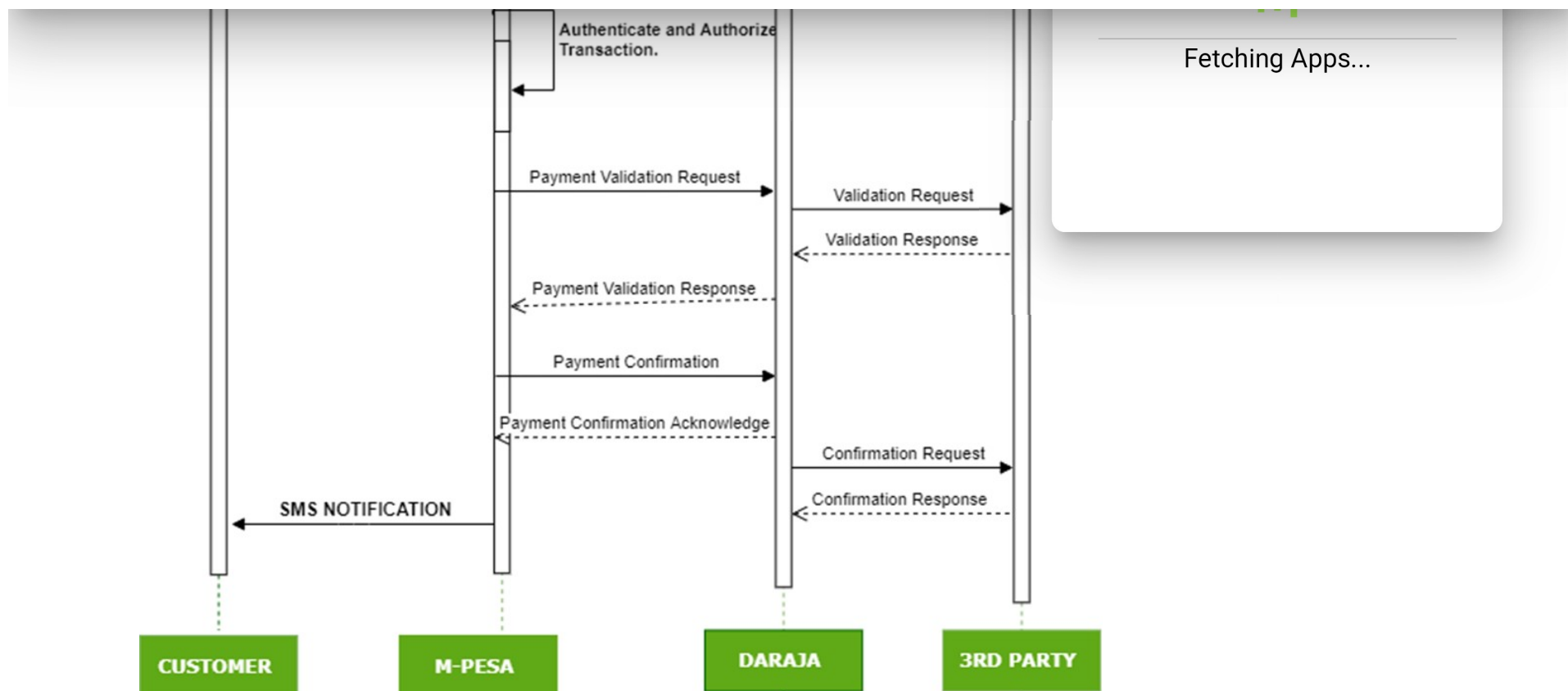
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The process is explained below:

1. Payment request to a pay bill by a customer from their phone.
2. M-PESA receives requests and validates them internally
3. M-Pesa then checks if **External Validation** has been activated for the paybill.
4. If External Validation is **enabled**:
 - a. M-PESA first sends a **Validation request** to the Validation URL registered in the system by the integrated partner with

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d. As explained in item **6** below, If you had registered a default action value to **completed**, M-Pesa sends a **Confirmation request** to your Confirmation URL with the details of the complete transaction. Thus you shall receive **two** API calls on your system.

e. If you had registered **cancelled** as the default action, M-Pesa simply cancels the transaction and no Confirmation request is sent. The transaction is then complete

5. If External Validation is **disabled**, M-Pesa automatically completes the transaction, and M-Pesa sends a **Confirmation** request to the Confirmation URL registered in the system. This is the **only** API call you shall receive on your end.
6. If External Validation is **enabled**, but for some reason, M-Pesa could **not** reach your endpoint to validate the transaction within the stipulated time period (usually < 8 seconds from the moment the request leaves M-PESA), or no response was received by the time M-PESA terminates the request, it checks on the default action value saved during registration of the URLs. If the default action was set to **Completed**, M-PESA automatically completes the transaction and also tries to send a Confirmation request to your other endpoint. If the default action was set to **Cancelled**, M-Pesa simply cancels the transaction and no Confirmation callbacks are sent. The transaction is then complete.
7. If no URLs are registered in the system, M-PESA automatically completes the request.
8. M-PESA then sends an SMS notification to both the customer and the paybill owner with the results of the transaction as usual.
9. When external notifications fail to be sent, you can use Pull transaction API for reconciliation or check on the M-PESA Org portal <https://org.ke.m-pesa.com/login.action> and cross-check against received callbacks. The portal has all the payments ever made available, whether you received the callback or not.

URL Requirements

1. Use publicly available (Internet-accessible) IP addresses or domain names.
2. All Production URLs must be HTTPS, on Sandbox you're allowed to simulate using HTTP.
3. Avoid using keywords such as M-PESA, M-Pesa, M-Pesa, Safaricom, exe, exec, cmd, SQL, query, or any of their variants in either upper or lower cases in your URLs.
4. **Do not** use public URL testers e.g. [ngrok](#), [mockbin](#), or [requestbin](#), **especially** on production, they are also usually blocked by the API. Use your own application URLs and do not make them public or share them with any of your peers.

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As you set up the default value, the words "Cancelled/Completed" must be in sentence

Request Body

```
{
  "ShortCode": "601426",
  "ResponseType": "[Cancelled/Completed]",
  "ConfirmationURL": "[confirmation URL]",
  "ValidationURL": "[validation URL]"
}
```

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Request Parameter Definition

Name	Description	Type	Sample Values
Validation URL	This is the URL that receives the validation request from the API upon payment submission. The validation URL is only called if the external validation on the registered shortcode is enabled. (By default External Validation is disabled).	URL	https://ip or domain:port/path
ConfirmationURL	This is the URL that receives the confirmation request from API upon payment completion.	URL	https://ip or domain:port/path
ResponseType	This parameter specifies what is to happen if for any reason the validation URL is not reachable. Note that, this is the default action value that determines what M-PESA will do in the scenario that your endpoint is unreachable or is unable to respond on time. Only two values are allowed: Completed or Cancelled. Completed	String	- Canceled - Completed

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Shortcode

Usually, a unique number is tagged to an M-PESA pay bill/till number of the organization.

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Response Body

```
{
  "OriginatorCoversationID": "7619-37765134-1",
  "ResponseCode": "0",
  "ResponseDescription": "success"
}
```

Response Parameters Definition

Name	Description	Type	Sample Values
OriginatorCoversationID	This is a global unique identifier for the transaction request returned by the API proxy upon successful request submission.	Alpha-numeric	An alpha-numeric string of fewer than 20 characters
ResponseCode	It indicates whether Mobile Money accepts the request or not.	Alpha-numeric	0
ResponseDescription	This is the status of the request.	String	Success

NB: Before any API request register URL is included, you will require authorization by generating an access token from the authorization API to allow you successfully make an API call.

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body below;

Result Body

Validation Request

The validation request is received only by partners who have the External Validation feature enabled. BuyGoods (Till Number) and require validating a payment before M-PESA completes the transaction. You will first receive a validation request to validate on your Validation URL.

The **Confirmation and Validation** result posted on your URLs after payment will have the structure below:

```
{
  "TransactionType": "Pay Bill",
  "TransID": "RKTQDM7W6S",
  "TransTime": "20191122063845",
  "TransAmount": "10",
  "BusinessShortCode": "600638",
  "BillRefNumber": "invoice008",
  "InvoiceNumber": "",
  "OrgAccountBalance": "",
  "ThirdPartyTransID": "",
  "MSISDN": "25470****149",
  "FirstName": "John",
  "MiddleName": "",
  "LastName": "Doe"
}
```

Validation Response

After receiving the validation request, you are required to process it and respond to the API call and inform M-PESA either to accept or reject the payment. To accept, you send the below JSON making sure the value of ResultCode is 0 (zero), but the

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To reject a transaction, you send the same JSON above, but with the ResultCode set as C2B00011 (see the error table below), BUT NOT 0. The ResultDesc should be set as Rejected as shown below

```
{
  "ResultCode": "C2B00011",
  "ResultDesc": "Rejected",
}
```

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Result Error Codes

ResultCode	ResultDesc
C2B00011	Invalid MSISDN
C2B00012	Invalid Account Number
C2B00013	Invalid Amount
C2B00014	Invalid KYC Details
C2B00015	Invalid Shortcode
C2B00016	Other Error

Results Parameter Definition

Name	Description	Type	Sample Values
TransactionType	The transaction type specified during the payment request.	String	Buy Goods or Pay Bill

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	YEAR+MONTH+DATE+HOUR+MINUTE+SECOND (YYYYMMDDHHMMSS) Each part should be at least two digits apart from the year which should be at least four digits.		
TransAmount	This is the amount transacted (normally a numeric value), money the customer pays to the Shortcode. Only whole numbers are supported.		
BusinessShortCode	This is the organization's shortcode (Paybill or Buygoods - a 5 to 6-digit account number) used to identify an organization and receive the transaction.	String	Shortcode (5 to 6 digits) e.g. 654321
BillRefNumber	This is the account number for which the customer is making the payment. This is only applicable to Customer PayBill Transactions	String	An alpha-numeric value of up to 20 characters.
OrgAccountBalance	The current utility account balance of the payment-receiving organization shortcode. For validation requests, this field is usually blank whereas, for the confirmation message, the value represents the new balance after the payment has been received.	Decimal	30671
ThirdPartyTransID	This is a transaction ID that the partner can use to identify the transaction. When a validation request is sent, the partner can respond with ThirdPartyTransID and this will be sent back with the Confirmation notification	String	1234567890
MSISDN	This is a masked mobile number of the customer making the payment.	Numeric	25470****149

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LastName

The customer's last name is as per the M-Pesa register. This parameter can be empty.

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NB: Data minimization is a requirement from the data protection office not to share customers' personal information without their consent.

Validation Response - Expected from your validation URL.

Name	Description	Type
ResultCode	A code indicating whether to complete the transaction. 0(Zero) always means complete. Other values mean canceling the transaction, which also determines the customer notification SMS type.	Alpha-numeric
ResultDesc	Short validation result description	String
ThirdPartyTransID	An optional value that can be used to identify the payment during a confirmation callback. If a value is set, it would be passed back in a confirmation callback.	String

Confirmation acknowledgment - Expected from your Confirmation URL.

Name	Description	Type
ResultCode	A code indicating the receipt status of the confirmation callback.	Number



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